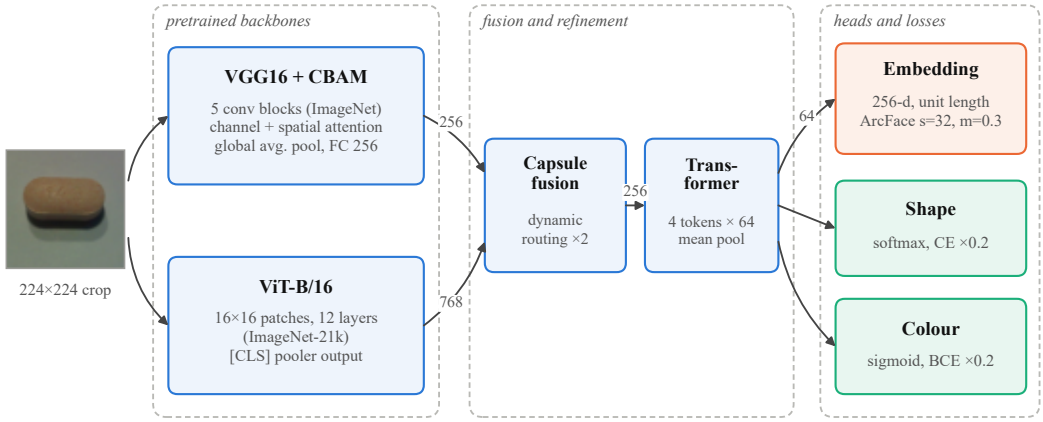


Figure 1: The hybrid network. VGG16 with CBAM attention and ViT-B/16 see the same crop; a capsule block fuses their features by dynamic routing and a transformer block refines the result into a unit-length 256-d embedding trained with ArcFace, with auxiliary shape and colour heads. Numbers on the arrows are feature sizes.

PharmaScan: Approach and Experiments (section preview)

Team

1 Approach

A detected pill is cropped, embedded by a hybrid CNN–ViT network, and compared by cosine similarity with the patient's enrolled medicines or with a product catalogue. Decision thresholds are calibrated on validation medicines only, and an OCR imprint reader adds a second, independent check.

1.1 Network

Figure 1 shows our network and Table 1 its layers. A VGG16 branch [Simonyan and Zisserman, 2015] with CBAM channel and spatial attention [Woo et al., 2018] and a ViT-B/16 branch [Dosovitskiy et al., 2021] see the same 224×224 crop: the CNN contributes local texture, colour and edges, the ViT global layout through self-attention between patches. A capsule block [Sabour et al., 2017] turns each branch into a 256-d capsule and fuses them by dynamic routing: the coupling coefficients start equal and move towards the branch that agrees with the fused vector. A small transformer encoder [Vaswani et al., 2017] splits the fused vector into four 64-d tokens and refines it; a linear head maps the mean token to the embedding $\mathbf{z} = f_\theta(x)$, with $\|\mathbf{z}\| = 1$ and 256 dimensions. Auxiliary heads predict the pill's shape and colours. All layers after the backbones use Glorot-uniform initialisation.

Preprint.

Table 1: The network's layers (inference mode). Pretrained weights: VGG16 on ImageNet, ViT-B/16 on ImageNet-21k.

Block	Details	Output
CNN branch	VGG16 conv. blocks → CBAM → global average pool → Dense, ReLU	256
ViT branch	ViT-B/16 (`vit-base-patch16-224-in21k`), pooler output	768
Capsule fusion	per branch: Dense 256, ReLU, dropout 0.4, Dense 256, batch norm; dynamic routing, 2 iterations, residual update	256
Refinement	4 tokens × 64 + learned positions; 2 post-norm encoder layers (4 heads, FFN 128, dropout 0.4); mean pool	64
Embedding head	Linear, ℓ_2 normalisation	256
Auxiliary heads	shape (softmax), colour (multi-label sigmoid)	–

Table 2: Training settings of the served model (run 4). Selection: top-1 on validation-enrolled medicines (best: phase 2, epoch 35). Seed 2026.

Setting	Phase 1	Phase 2
Trainable	all but ViT, VGG blocks 1–3	all
Max. epochs; batch ($P \times K$)	25; 32×4	35; 16×4
LR: backbones / new layers / ArcFace	3×10^{-5} / 3×10^{-5} / 10^{-3}	10^{-5} / 10^{-5} / 10^{-4}
Optimiser	AdamW, weight decay 10^{-3}	same
LR schedule	reduce on plateau ($\times0.5$, patience 3)	same
Early stopping	patience 6, best weights restored	same

1.2 Training

Objective. The embedding is trained with ArcFace [Deng et al., 2019] over the 1,710 training appearance groups (pill types that share imprint, shape, colour and size form one group), plus auxiliary losses:

$$\mathcal{L} = -\log \frac{e^{s\cos(\theta_y+m)}}{e^{s\cos(\theta_y+m)} + \sum_{j\neq y} e^{s\cos\theta_j}} + \lambda_s \mathcal{L}_{\text{shape}} + \lambda_c \mathcal{L}_{\text{colour}}, \tag{1}$$

with $s = 32$, $m = 0.3$ and $\lambda_s = \lambda_c = 0.2$, where θ_j is the angle between $\mathbf{z}$ and the j-th class weight, $\mathcal{L}_{\text{shape}}$ is a cross-entropy and $\mathcal{L}_{\text{colour}}$ a binary cross-entropy (unknown labels ignored). The margin warms up over the first three epochs $(0, 0.1, 0.2)$, then stays at 0.3.

Optimisation. Table 2 lists the settings. Training has two phases: in phase 1 the ViT and VGG blocks 1–3 are frozen; in phase 2 everything trains. Batches hold P groups $\times$ $K{=}4$ photos each [Hermans et al., 2017]. AdamW [Loshchilov and Hutter, 2019] uses three learning-rate groups: pretrained backbones, new layers, and the randomly initialised ArcFace class table. These rates were the decisive choice. Our first run trained the new layers at 10^{-3} and its embedding collapsed to about 3 effective dimensions[1], enough to encode colour and shape but not imprints. Lower rates (3×10^{-5}, then 10^{-5}), a faster rate for the class table and the margin warm-up fixed it (§2.3, Figure 6).

Augmentation. Colour jitter is kept small (saturation 0.05, hue 0.0125) because colour identifies pills. To mimic detector error the crop window is jittered ($\times0.75$–1.35, centre shift $\leq 10\%$). For other phones and lighting we add brightness and contrast (±0.3), white balance ($\pm8\%$), downscaling to 40–160 px ($p{=}0.3$), JPEG quality 30–90 ($p{=}0.3$) and Gaussian noise ($\sigma{=}0.03$, $p{=}0.2$), and paste a neighbouring pill partly into the frame ($p{=}0.2$).

[1] exp of the entropy of the share of variance per principal direction: 256 if all directions are used equally, 1 for a line.

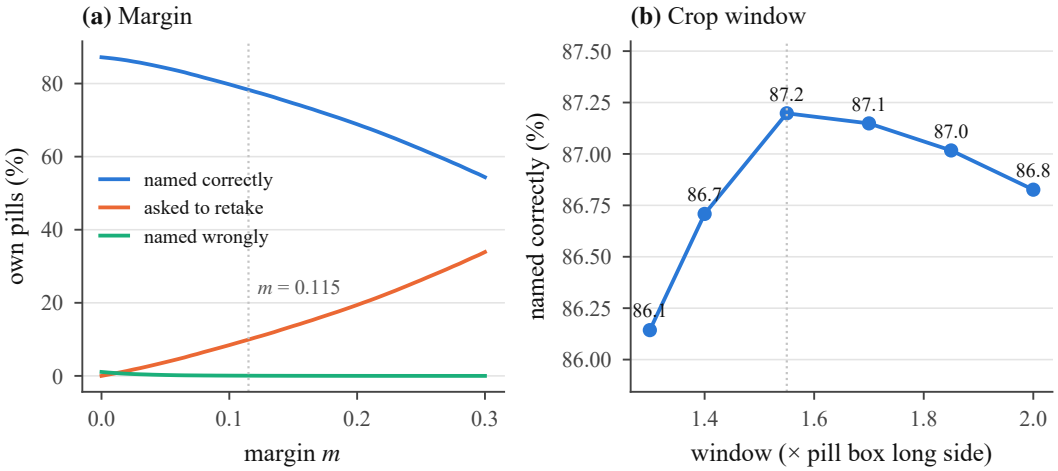


Figure 2: Settings chosen on validation (1,000 simulated patients, 10 medicines, 3 enrolment photos; τ rejects 95% of other medicines). (a) The margin trades correct names for retakes while wrong names stay near zero. (b) The crop window (note the narrow y-range).

1.3 Open-set decision rule and calibration

A patient enrols each medicine j from k=3 photos; its prototype is $\mathbf{p}_j = \sum_i f_\theta(x_{j,i}) / \|\sum_i f_\theta(x_{j,i})\|$. For a query with similarities $s_j = \mathbf{z}^\top \mathbf{p}_j$, sorted $s_{(1)} \geq s_{(2)}$,

$$\hat{y} = \begin{cases} \text{UNKNOWN} & s_{(1)} < \tau, \\ \text{RETAKE} & s_{(1)} \geq \tau \text{ and } s_{(1)} - s_{(2)} < m, \\ \arg\max_j s_j & \text{otherwise,} \end{cases} \tag{2}$$

where RETAKE asks the patient to turn the pill over and photograph it again. We calibrate on validation only, by simulating 1,000 patients with 10 medicines each. $\tau = 0.895$ rejects 95% of medicines that are not the patient's. As Figure 2a shows, the margin trades correct names for retakes while wrong names are already near zero; we take the largest margin with at most 10% retakes, $m = 0.115$. The crop window, $1.55\times$ the detected box's long side, is the best of a sweep from 1.3 to 2.0, although all windows lie within one point (Figure 2b).

1.4 Identification with imprints and calibrated confidence

A pill that is not in the gallery is ranked against 4,936 US catalogue products; a product's look score s is the similarity of its closest reference photo. Florence-2-large [Xiao et al., 2024] reads the imprint zero-shot at four rotations (Florence-2-base on the CPU path). Each recorded imprint token gets an IDF weight in $[0, 1]$, so a rare token ('EC5') counts a lot and a common one ('93', printed on hundreds of Teva pills) little. A product's imprint score r is the weight of its best token found in the reads, and the top K=100 products by look are re-ranked by $s + \alpha r$ with $\alpha = 0.05$, both chosen on validation (Figure 7b). In the patient's gallery, only tokens that distinguish the enrolled imprints from each other confirm or contradict a match. Raw similarities are not probabilities (pills that are not in the catalogue still score about 0.93), so a logistic model fitted on validation, in the spirit of Platt scaling [Platt, 1999], gives the stated confidence $\sigma(-12.50 + 12.50\, s_1 + 72.13\,(s_1 - s_2) + 3.92\, r_1)$. Here s_1, s_2 are the look scores of the top two re-ranked products and r_1 is the imprint score of the first.

1.5 Implementation and reproducibility

We use PyTorch [Paszke et al., 2019] 2.11 and Hugging Face Transformers 5.19, and train on one NVIDIA RTX 4080 SUPER (16 GB). The model is served as a FastAPI service on a Hugging Face ZeroGPU Space, with 92 automated tests. Each stage is one script in the supplementary code (`make_splits`, `train_stage2`, `calibrate_stage2`, `evaluate_stage2`,

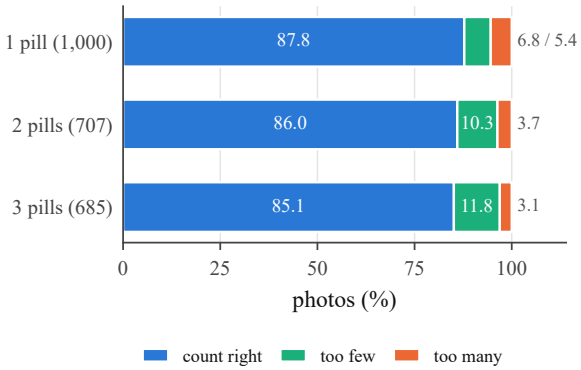


Figure 3: The pill finder on C3PI photos with known pill counts (2- and 3-pill photos composed from single-pill photos). Small shares are listed after each bar in legend order.

`tune_identify_rerank`, `calibrate_identify`, `make_paper_figures`), and every number reported here is written to a JSON file by those scripts.

1.6 Software 3.0 contribution

We used an AI coding agent, Claude Code [Anthropic, 2026], throughout, programming in natural language in the sense of Software 3.0 [Karpathy, 2025]. It wrote the model, training, calibration and serving code and the evaluation scripts from our instructions; we reviewed and ran them. The effective-dimension measure in the training log located run 1's collapse in the new layers' learning rate, and the fix was proposed and tested in the same workflow. Every proposal was adopted or rejected on validation measurements, not on the agent's word: two imprint-reader fine-tunes it set up were rejected (Table 6).

2 Experiments

2.1 Setup

Test sets. No test medicine appears in training in any form, and every setting is fixed on validation before a test set is scored, once. *C3PI held-out*: 4,005 photos of 200 appearance groups (the medicines of ePillID's official holdout fold and their look-alikes), photographed like the training data. *MEDISEG* [MEDISEG]: 13,973 pills of 32 Hong Kong products, taken with other phones and in cluttered scenes; median pill size 81 px. *ePillID* [Usuyama et al., 2020]: the published benchmark (fold 4, two-sided queries against references of all 4,902 types).

Healthcare-relevant metrics. We simulate patients with 10 enrolled medicines (3 photos each). For the patient's own pills we report the share *named correctly*, *named wrongly* (the safety-critical error), *retake* and *called unknown*; for pills that are not theirs, the share *rejected*. A wrong name may be acted on, whereas "unknown" only costs time, so wrong names matter most. We also report top-k accuracy, AUROC, ePillID's MAP and GAP, and the reliability of the stated confidence. 95% intervals come from resampling medicines.

Baselines. (i) Run 1: the same network with a common recipe (new layers at 10^{-3}, no margin warm-up). (ii) The published ePillID models [Usuyama et al., 2020]. (iii) Look only versus look plus imprint, for the gallery and the catalogue. (iv) For the pill finder that feeds the model, an earlier finder that returned exactly one box on only 28% of single-pill photos; ours counts 85–88% of 1-, 2- and 3-pill photos right (Figure 3).

2.2 Main results

Table 3 gives the test results and Figure 4 the three headline measures. In the conditions it was trained for, the served model (run 4) names 76.0% of a patient's own pills correctly and only 0.04%

Table 3: Test results on medicines never seen in training (%; 10-medicine galleries, 3 enrolment photos, calibrated τ and m). Bottom block: the served pipeline with imprint reading (GPU reader; CPU reader in brackets).

	Run 2	Run 3	Run 4 (served)
C3PI held-out: top-1 among 200 medicines	82.0	**84.4**	83.2
gallery: named correctly / wrongly	78.6 / 0.1	**80.4** / 0.0	76.0 / 0.04
other medicines rejected	94.2	94.1	**94.8**
MEDISEG: top-1 among 32 products	38.9	41.8	**48.2**
gallery: named correctly / wrongly	17.6 / 1.1	18.7 / 1.0	18.7 / **0.7**
own called unknown / other rejected	51.5 / 78.7	64.2 / 89.3	70.4 / **93.1**
ePillID strict (no test type in training): MAP@1	13.7	37.1	**42.3**
C3PI gallery + imprint check: named correctly / wrongly			78.3 (77.3) / 0.06
retake / called unknown			7.2 / 14.5
Catalogue (4,936 products) + imprint: top-1 / top-5			80.2 (79.2) / 94.4
look only: top-1 / top-5			71.6 / 91.7

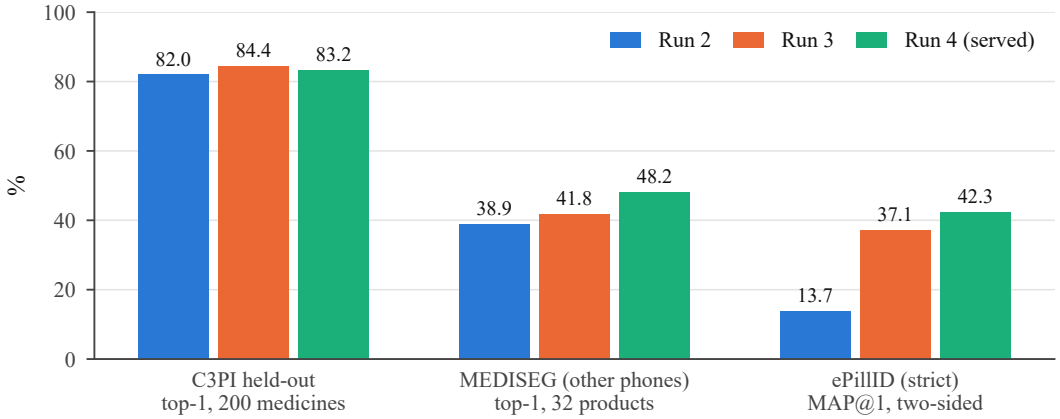


Figure 4: Test results on medicines never seen in training, each scored once. Run 3 adds C3PI studio reference photos; run 4 adds augmentation for other phones and jittered crops.

wrongly (95% CI 0.02–0.07%), while rejecting 94.8% of pills that are not the patient's (Figure 5a). The imprint check raises correct names to 78.3% (wrong 0.06%) by settling retakes. Against the catalogue, the right product comes first for 80.2% of held-out pills and is in the top 5 for 94.4%, against 71.6% and 91.7% by look alone. Adding reference photos (run 3) nearly tripled ePillID MAP@1 (13.7→37.1%), and robustness augmentation (run 4) gained 6.4 points on MEDISEG at a small cost in C3PI conditions; run 4 is the served model.

2.3 Ablations and hyperparameters

Table 4 isolates each design decision. The learning rates matter most. With the new layers at 10^{-3} the embedding collapses to about 3 effective dimensions and validation top-1 stalls at 77%, while lower rates keep 12–15 dimensions and reach 98% (Figure 6). The decision settings are smooth (Figure 2): the crop window moves correct names by under one point, and the margin trades correct names for retakes almost one for one. The imprint weight has a clear optimum (Figure 7b), because a large α lets false OCR reads promote wrong products; the imprint helps most when the pill's own imprint is actually read (Figure 7a).

2.4 Robustness

Other phones are the main weakness (Table 3, Figure 5). On MEDISEG, top-1 drops to 48% and the C3PI-calibrated threshold calls 70% of own pills unknown, though wrong names stay below 1%: the system fails safe but is not yet usable there. Small pills are the main factor (Figure 8a: 36–37% top-1 under 100 px, 62% above), and more enrolment photos help only partly (Figure 8b). *Pills outside the*

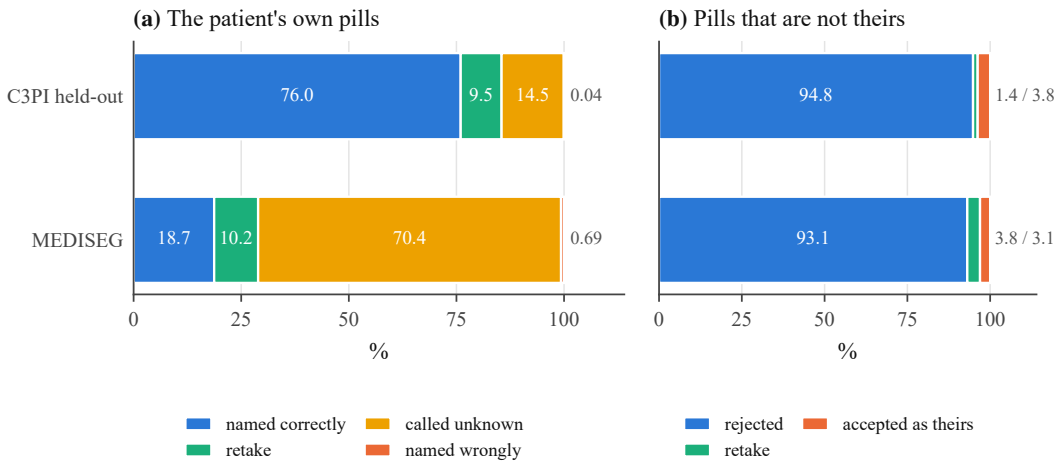


Figure 5: What the patient sees (run 4, embedding only; 10 medicines, 3 enrolment photos). Small shares are listed after each bar in legend order. In C3PI conditions most own pills are named correctly and wrong names are rare; on other phones the threshold transfers badly and most own pills are called unknown, which is safe but unhelpful.

Table 4: Ablations (validation unless marked *test*).

Change	Effect	Kept
Run 1→2: lower LRs (3×10^{-5}, 10^{-5}), separate ArcFace LR, margin warm-up	top-1 in 10 medicines 77.2→98.3%; AUROC 0.78→0.981; correct at 95% rejection 17→90.7%; dims ≈3→12–15	yes
Run 2→3: + studio reference photos	validation 98.1%; *test* ePillID MAP@1 13.7→37.1%	yes
Run 3→4: + photo and crop-jitter augmentation	*test* MEDISEG 41.8→48.2%; C3PI 84.4→83.2%	yes
Crop window 1.3–2.0×	named correctly 86.1–87.2%	1.55
Margin m 0–0.3	named correctly 87→54%; retakes 0→34%	0.115
Imprint re-ranking α, K	top-1 72.1% (α=0)→81.6%; *test* 71.6→80.2%	0.05, 100
Gallery check: distinct tokens only	named wrongly 35→30; other pills accepted 6,158→6,150	yes
Official ePillID protocol (runs 5–7)	ePillID MAP 0.57–0.60; MEDISEG 52–54%	benchmark

catalogue: 69.8% of MEDISEG pills get *no confident match* and 7.0% *possible match*, while 75.8% of catalogue pills get *possible match*, and the stated confidence tracks observed accuracy (Figure 9). *CPU deployment*: with the smaller reader, identification loses one point (79.2%) and wrong names do not rise (Table 3).

2.5 Comparison with the ePillID benchmark

For a like-for-like comparison, runs 5–7 follow ePillID's protocol and train on the references of all types (Table 5, Figure 10). Our embedding alone (MAP 0.57–0.60) is in the range of the paper's plain classifiers, not its best model. Re-ranking by imprint, which the paper's model did not use, raises MAP to 0.77–0.78 and GAP@1 to 0.921, above the paper's 0.912, while MAP@1 stays lower (69.0 vs. 92.0%). The paper's ablations attribute its advantage to its training recipe (contrastive and hard-mined triplet losses with softmax and ArcFace over side-specific classes, a 2,048-d embedding) rather than its backbone. We ruled out our crop style, the 64-d pooling, and local versus global pooling as causes.

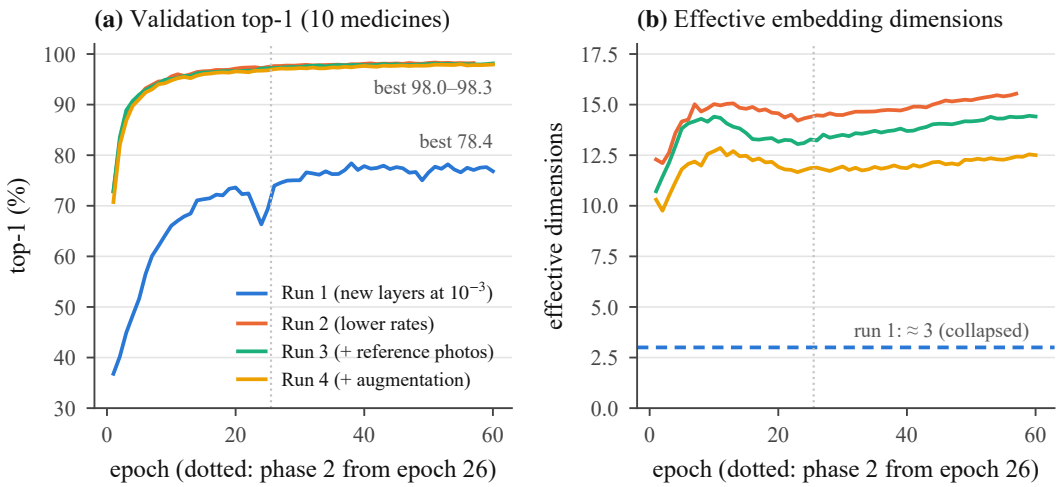


Figure 6: Training. (a) Validation top-1 within a gallery of 10 medicines never trained on. (b) Effective embedding dimensions: with the new layers at 10^{-3} the embedding collapsed to about 3.

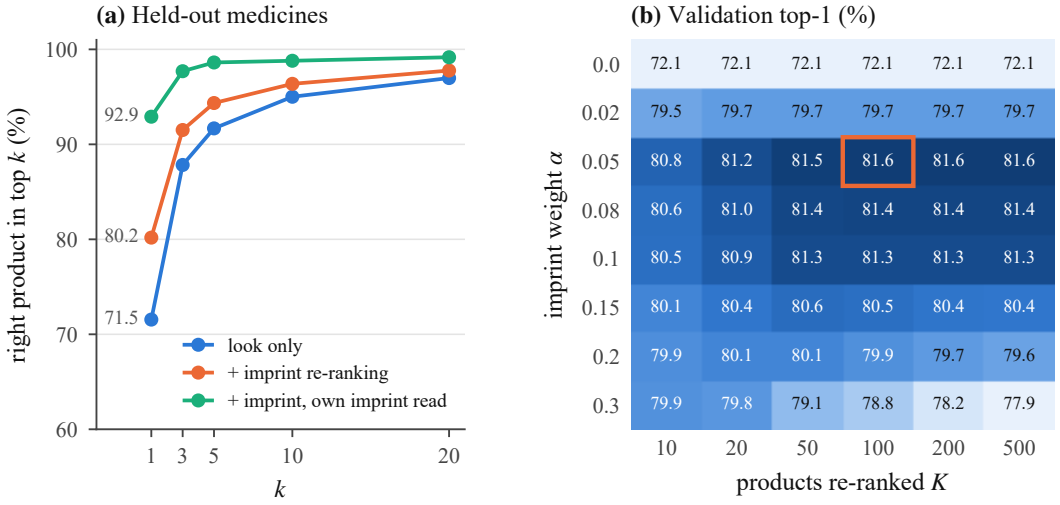


Figure 7: Catalogue identification. (a) Top-k on held-out medicines: the imprint helps most when the pill's own imprint is read (92.9% first vs. 68.6% when it is not). (b) Validation grid for α and K; the box marks the served setting.

2.6 Error analysis and interpretation

Imprints. The reader finds a token of the pill's own imprint on about 47% of validation photos. Figure 11 shows where it fails: debossed, small and round pills are hardest, and C3PI's own raters marked many imprints unreadable.

Look-alikes. The hardest errors are medicines that look the same (Figure 12). With two strengths of one ingredient in a gallery, one is named as the other 0.11% of the time, or 0.58% when shape and colour also match. The imprint check settles retakes but cannot confirm strength, so strength is taken from the pharmacy label. Same-manufacturer pairs that share an imprint prefix are confused 0.01% of the time overall, but the worst pairs share Teva's 'TV', e.g. tramadol 50 mg (TV;58) and carvedilol 12.5 mg (TV;7295) at 5.8%.

What the model learned. Photos of a medicine never seen in training form tight clusters (Figure 14), and the calibrated threshold separates a photo's similarity to its own medicine from its

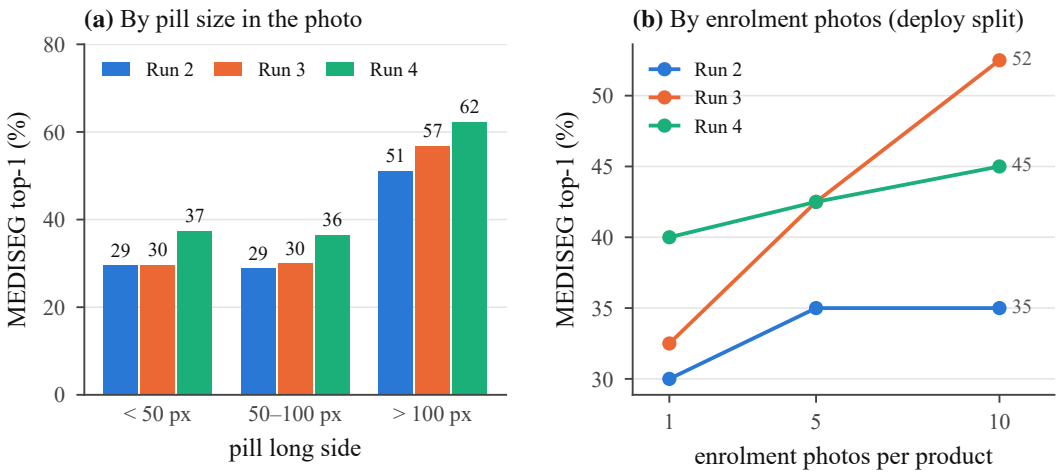


Figure 8: Robustness on MEDISEG (32 Hong Kong products, other phones, none in training). (a) Top-1 by pill size in the photo. (b) MEDISEG's deploy split (40 query pills): more enrolment photos help only partly.

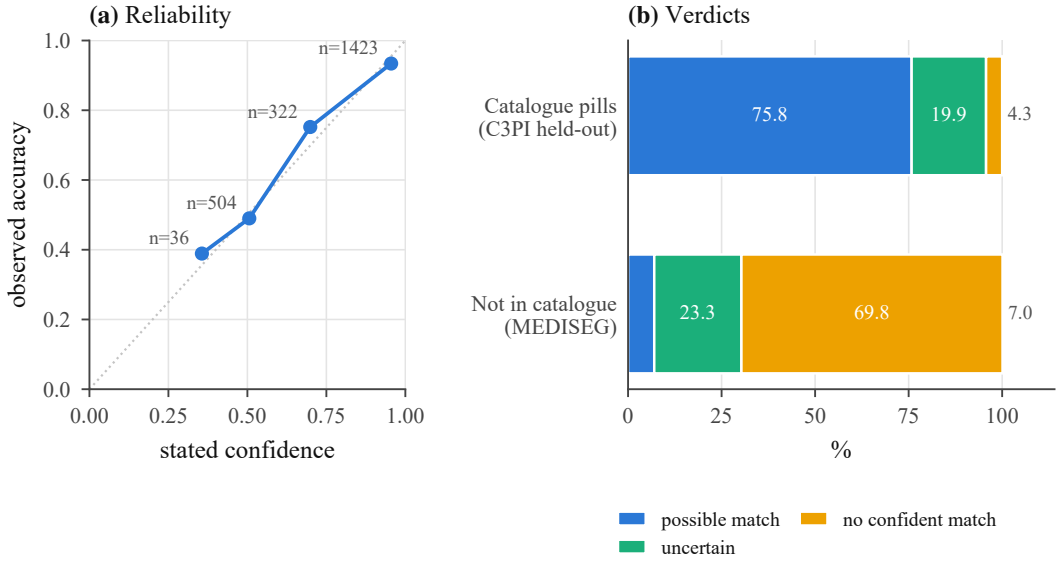


Figure 9: Calibrated confidence for identification. (a) Stated confidence against observed accuracy on held-out medicines. (b) Verdicts for catalogue pills and for pills not in the catalogue (MEDISEG).

similarity to the closest of nine others (Figure 13). Tracing the served model on single photos shows two things. Capsule routing moves from equal weights to about 9% CNN and 91% ViT, and the refinement block's token attention is nearly uniform, consistent with that block not being a bottleneck. Overall, the hybrid learns an embedding that transfers to unseen medicines when it is trained with the right learning rates. Calibration gives predictable behaviour on new medicines in familiar conditions but not across phones, and the imprint is evidence that complements the look.

Negative results. Table 6 lists what did not work. Two LoRA [Hu et al., 2022] fine-tunes of the imprint reader read more imprints on validation (44.0→53.5% and 45.0→50.7%). The first wrote memorised imprints onto new pills; the second doubled wrong names in galleries (30→61). Both were rejected, as were look-alike-character matching and per-candidate imprint scoring.

Table 5: ePillID, official protocol (fold 4, two-sided). The best published model is a ResNet-152 with compact bilinear pooling and multi-head metric learning. "+ imprint" uses OCR information the published models did not use.

	MAP	GAP	MAP@1	GAP@1	top-5
Usuyama et al. [2020], best	**0.958**	**0.812**	**92.0**	0.912	–
Usuyama et al. [2020], plain classifiers	.547–.727	.241–.448	35.7–56.8	.331–.551	–
Ours, run 5	0.597	0.207	45.7	0.526	77.5
Ours, run 5 + imprint	0.780	0.598	69.0	**0.921**	89.6
Ours, run 7 + imprint	0.774	0.614	67.6	0.916	89.5

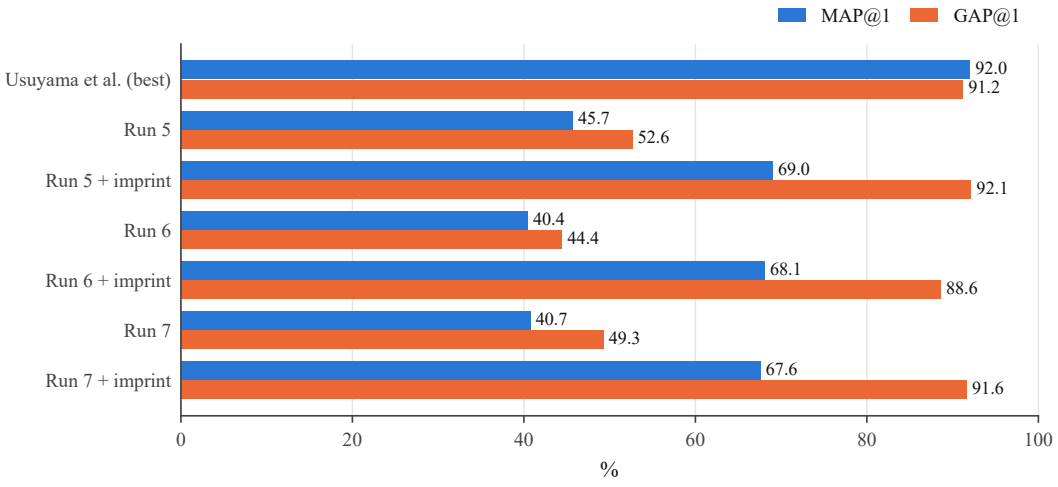


Figure 10: ePillID benchmark (official protocol): MAP@1 and GAP@1 of our runs with and without imprint re-ranking against the best published model.

2.7 Software 3.0 in coding and evaluation

The evaluation harness was written with the coding agent from natural-language specifications: the patient simulation, the look-alike and same-manufacturer stress tests, the top-k and confidence evaluations, the figures, and the API's 92 tests. Tests and review caught real defects, among them a safety bug in the label reader that linked an extended-release sticker to the immediate-release product; we fixed it by enforcing the release form. Our safeguards against over-trusting generated code were procedural: settings were chosen on validation, each test set was scored once, and every number in this paper is read from a file written by a script.

References

Anthropic. Claude code. `https://claude.com/claude-code`, 2026. The AI coding assistant used in this project.

Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. ArcFace: Additive angular margin loss for deep face recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.

Alexander Hermans, Lucas Beyer, and Bastian Leibe. In defense of the triplet loss for person re-identification. *arXiv preprint arXiv:1703.07737*, 2017.

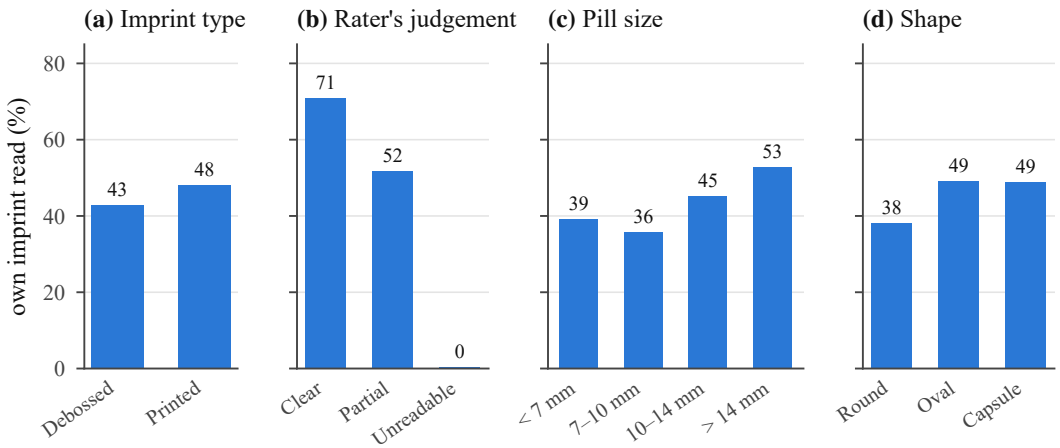


Figure 11: Share of validation photos on which a token of the pill's own imprint is read (Florence-2-large, zero-shot), by imprint type, C3PI rater's judgement of visibility, pill size and shape.

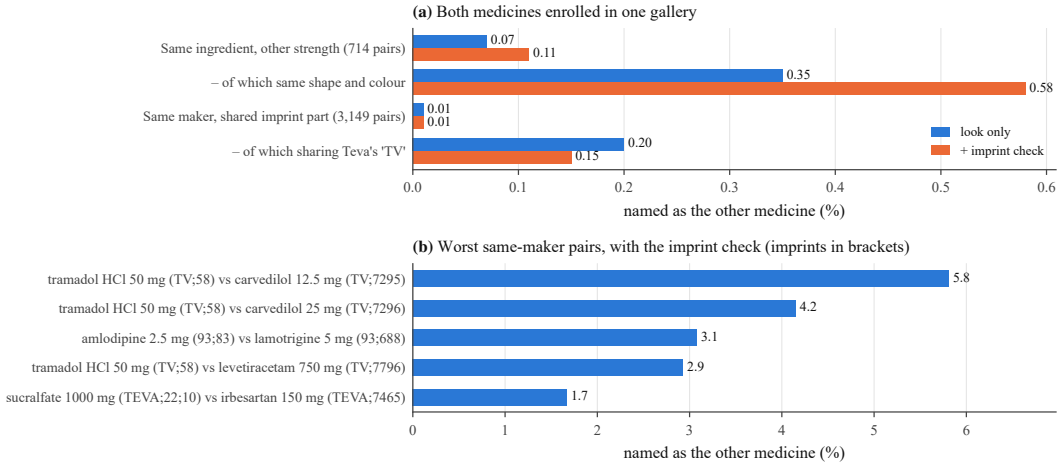


Figure 12: Look-alikes, with both medicines enrolled in one gallery. (a) Wrong-name rates by kind of pair, look only and with the imprint check. (b) The worst same-manufacturer pairs.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.

Andrej Karpathy. Software is changing (again). Keynote, Y Combinator AI Startup School, San Francisco, 2025. Introduces the term Software 3.0; VERIFY the URL of the recording.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.

MEDISEG. MEDISEG: a dataset of medication pill images. City, University of London figshare, `https://city.figshare.com/articles/dataset/MEDISEG/28574786`, 2026. CC BY 4.0. VERIFY authors and exact title from the figshare page; cite their paper if one exists.

Adam Paszke et al. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

John C. Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*. MIT Press, 1999.

Sara Sabour, Nicholas Frosst, and Geoffrey E. Hinton. Dynamic routing between capsules. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

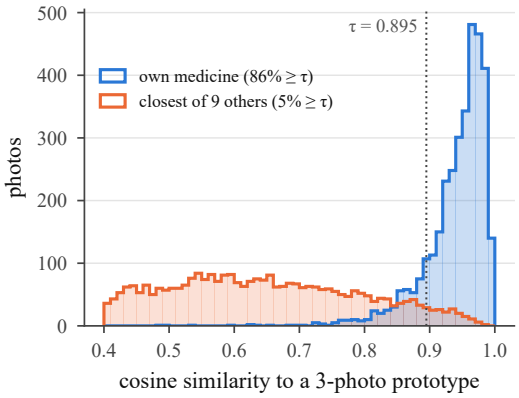


Figure 13: Held-out photos: similarity to their own medicine's 3-photo prototype and to the closest of 9 other medicines, with the threshold τ.

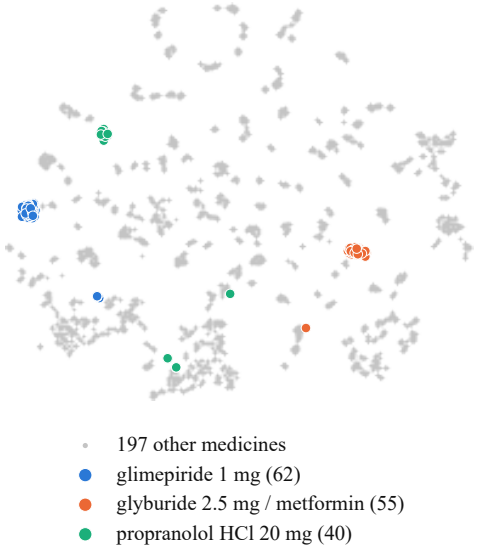


Figure 14: t-SNE [van der Maaten and Hinton, 2008] of the 4,005 held-out photos' embeddings; three medicines highlighted (photo counts in brackets).

Table 6: What did not work (validation).

Tried	Result	Decision
LoRA fine-tune of Florence-2-large, v1 (19,312 photos of training medicines)	imprints read 44.0→53.5%, but memorised imprints written onto new pills; identification and gallery check worse	rejected
LoRA v2 (also learns to output nothing on unreadable photos)	own imprint read 45.0→50.7%; identification 82.0→83.0%; gallery wrong names 30→61	rejected
Look-alike characters (O/0, B/8) treated as equal	measured and dropped	rejected
Scoring each candidate's imprint separately	measured and dropped	rejected
Runs 6–7: stronger fine-tuning; benchmark-aligned training	better validation and MEDISEG, not ePillID	not served

Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition. In *International Conference on Learning Representations (ICLR)*, 2015.

Naoto Usuyama, Natalia Larios Delgado, Amanda K. Hall, and Jessica Lundin. ePillID dataset: A low-shot fine-grained benchmark for pill identification. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2020.

Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9:2579–2605, 2008.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Sanghyun Woo, Jongchan Park, Joon-Young Lee, and In So Kweon. CBAM: Convolutional block attention module. In *European Conference on Computer Vision (ECCV)*, 2018.

Bin Xiao, Haiping Wu, Weijian Xu, Xiyang Dai, Houdong Hu, Yumao Lu, Michael Zeng, Ce Liu, and Lu Yuan. Florence-2: Advancing a unified representation for a variety of vision tasks. In

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.